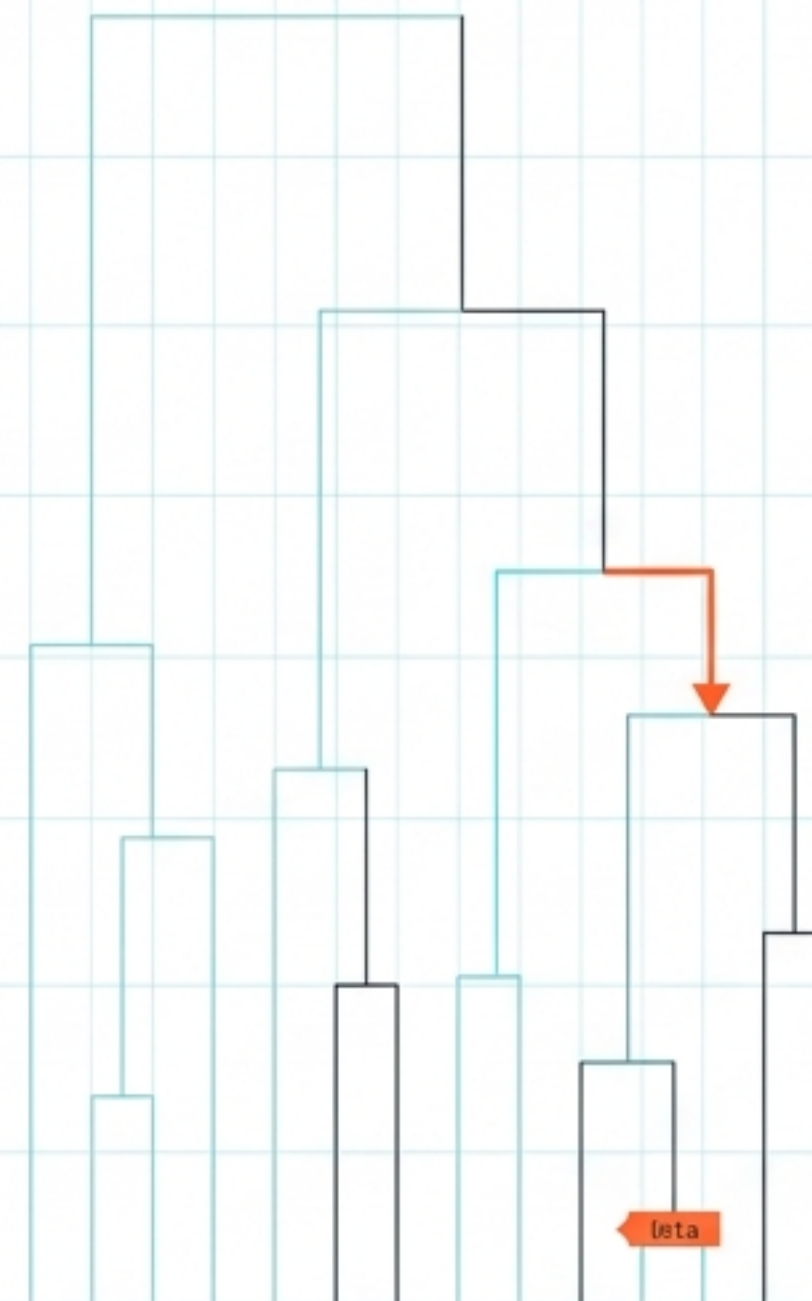


# The EDA Playbook

## Mastering Bivariate & Multivariate Analysis

Distilling narratives from raw data using  
Pandas and Seaborn.





# The Core Framework of EDA

Great exploratory data analysis is not about guessing which chart looks best. It is about matching the correct visual tool to the structural types of your data.

What are the data types  
of the columns?

[NUM] × [NUM]

Numerical vs Numerical

[NUM] × [CAT]

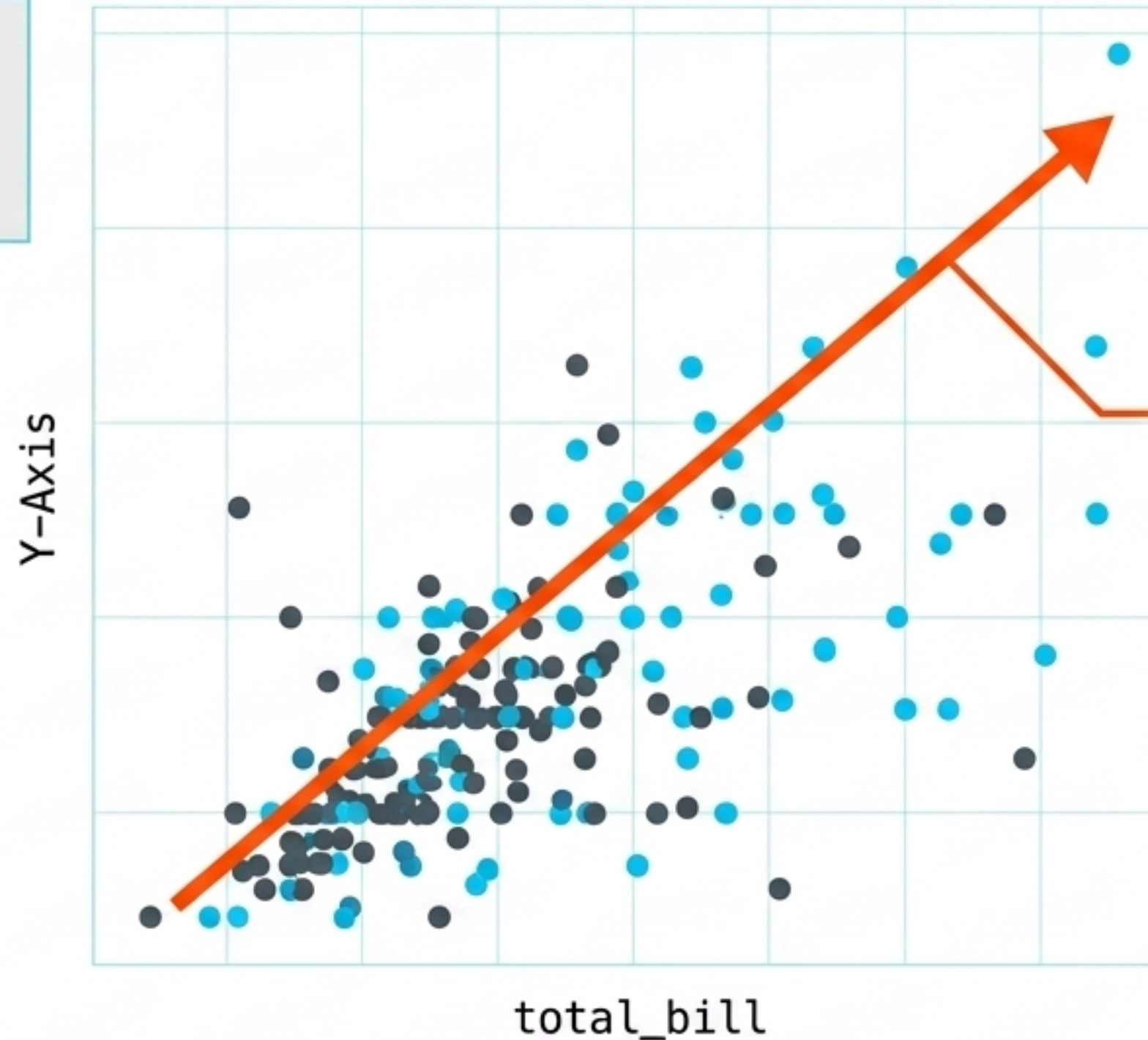
Numerical vs Categorical

[CAT] × [CAT]

Categorical vs Categorical

# Uncovering Correlations: The Scatter Plot

```
sns.scatterplot(  
    x='total_bill',  
    y='tip',  
    data=tips)
```

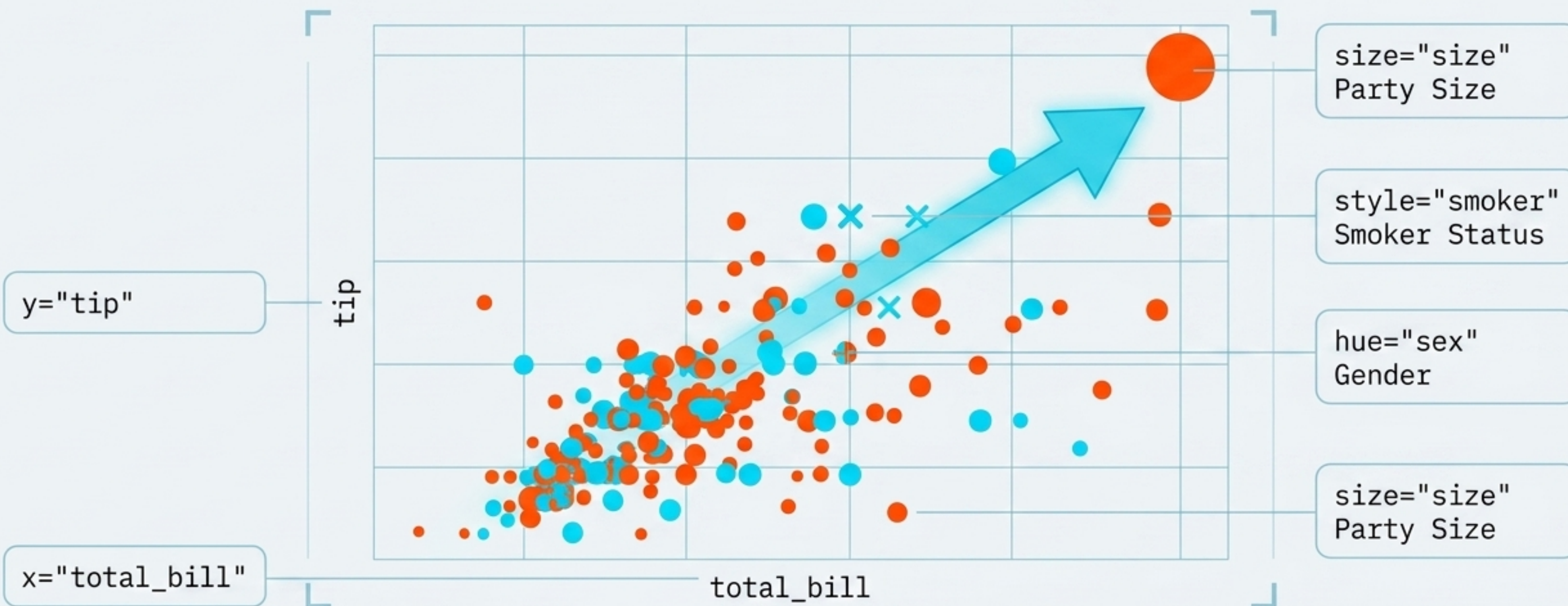


Higher bills logically dictate higher tips. The scatter plot establishes the baseline linear relationship between two continuous numerical values.



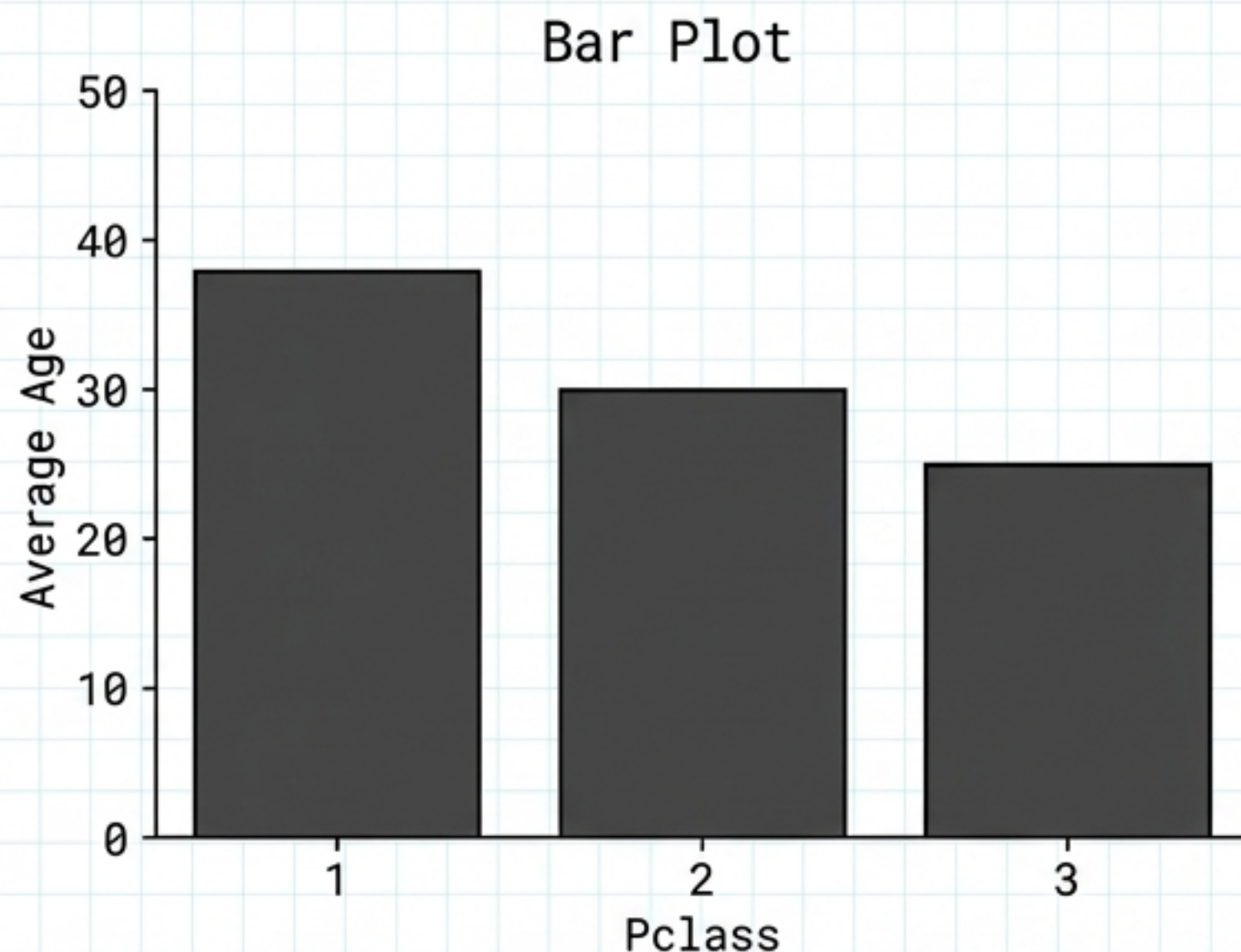
# Levelling Up: The 5-Dimensional Scatter Plot

Five distinct pieces of information packed into a single coordinate space. Multivariate analysis overlays categorical and numerical modifiers onto the base relationship.



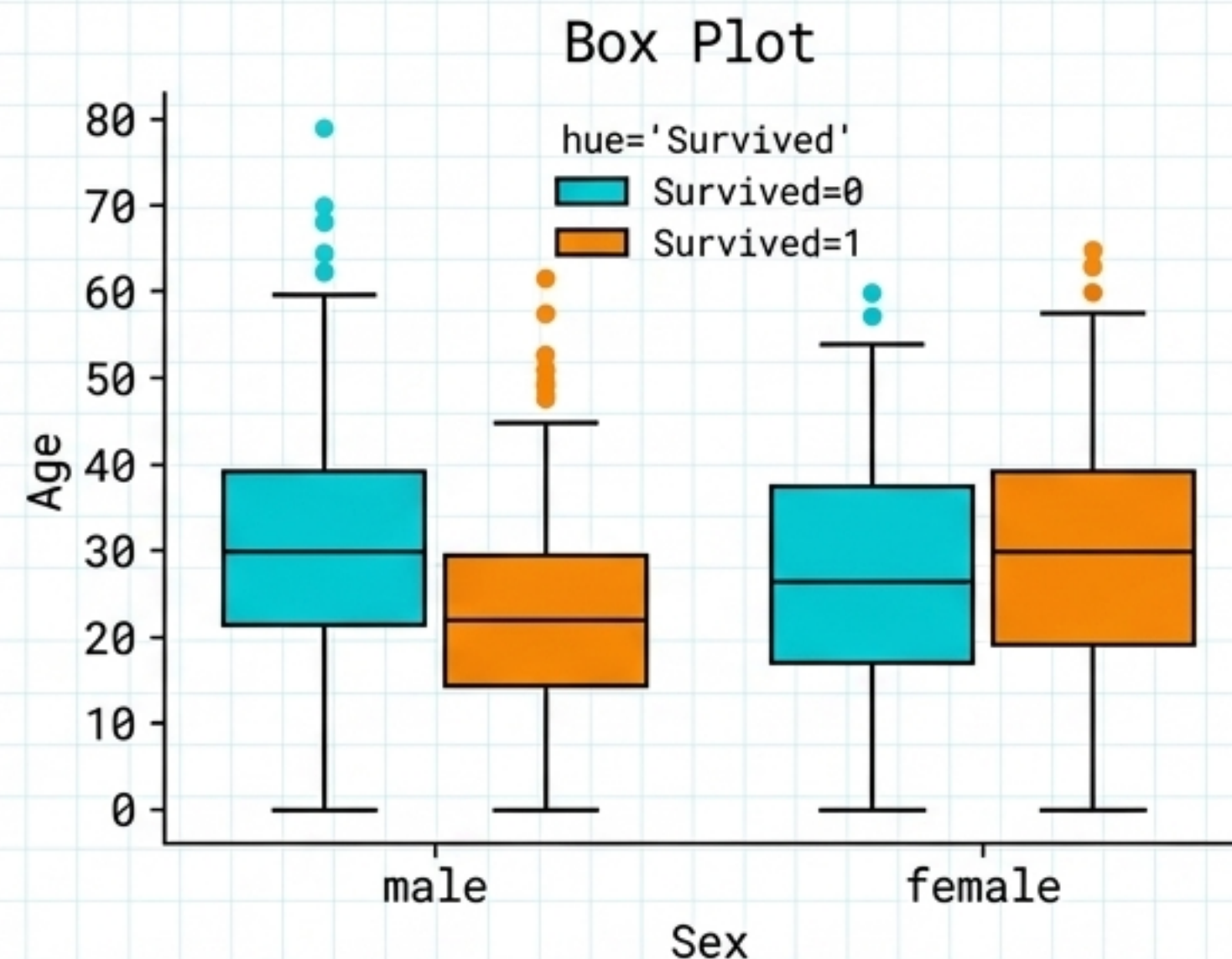


# Comparing Distributions Across Categories



Bar Plots isolate the mean.

**Observation:** Class 1 passengers were noticeably older on average (approx. 38 years) than Class 3 passengers (approx. 25 years).



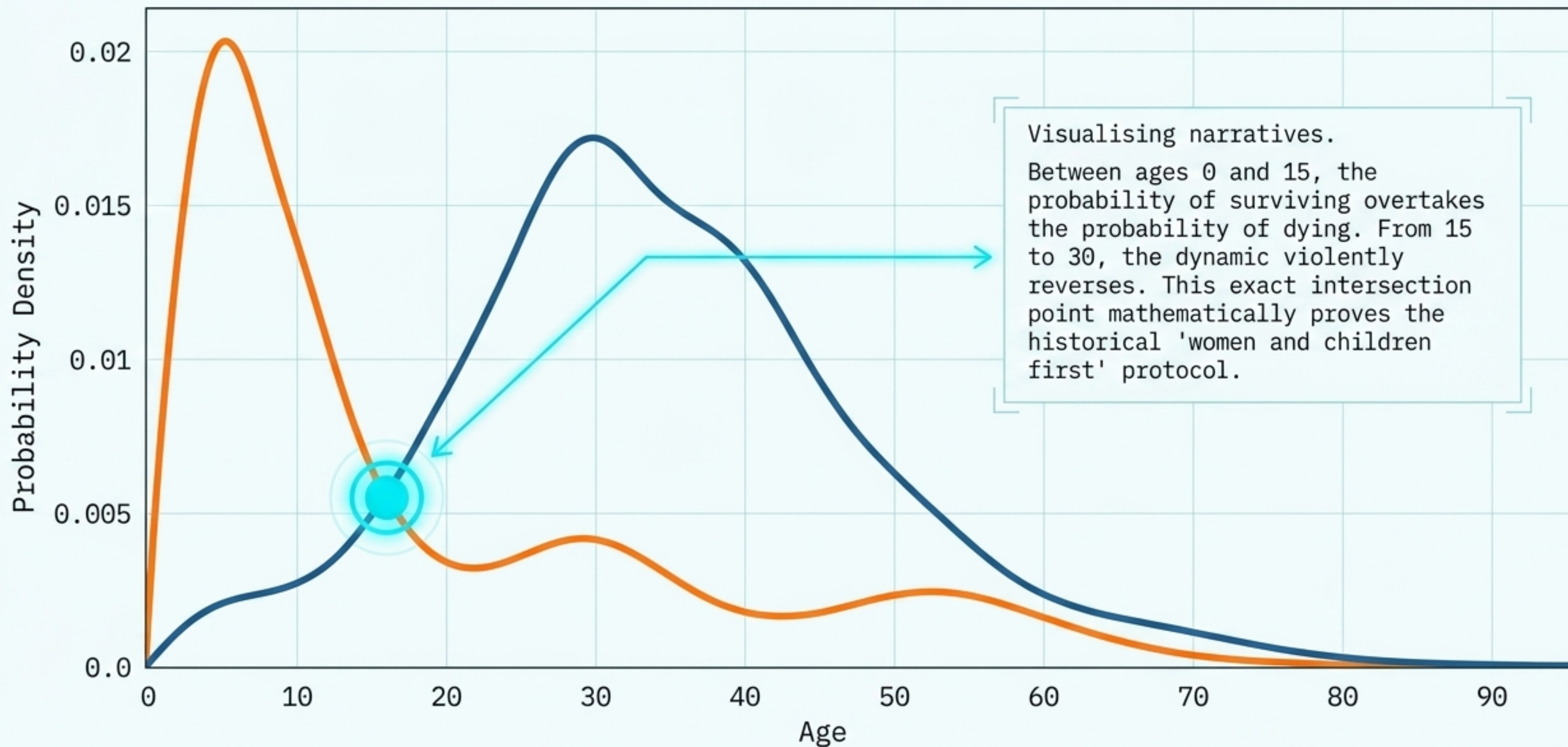
Box Plots reveal the spread and outliers.

**Observation:** Adding hue exposes that male survivors were heavily concentrated in the younger demographics, whereas female survivors skewed slightly older.



# The 'Aha!' Moment: Overlapping Density

[NUM] × [CAT]





# From Crosstabs to Visual Heat

Human eyes process colour intensity instantly. A heatmap translates raw categorical frequency into immediate visual weight, making the catastrophic death toll in Class 3 instantly obvious without reading a single number.

Raw Data: pd.crosstab Matrix

| Pclass | Survived |     |
|--------|----------|-----|
|        | 0        | 1   |
| 1      | 80       | 136 |
| 2      | 97       | 87  |
| 3      | 372      | 119 |



Visual Heat: Seaborn Heatmap

| Pclass | Survived |     |
|--------|----------|-----|
|        | 0        | 1   |
| 1      | 80       | 136 |
| 2      | 97       | 87  |
| 3      | 372      | 119 |

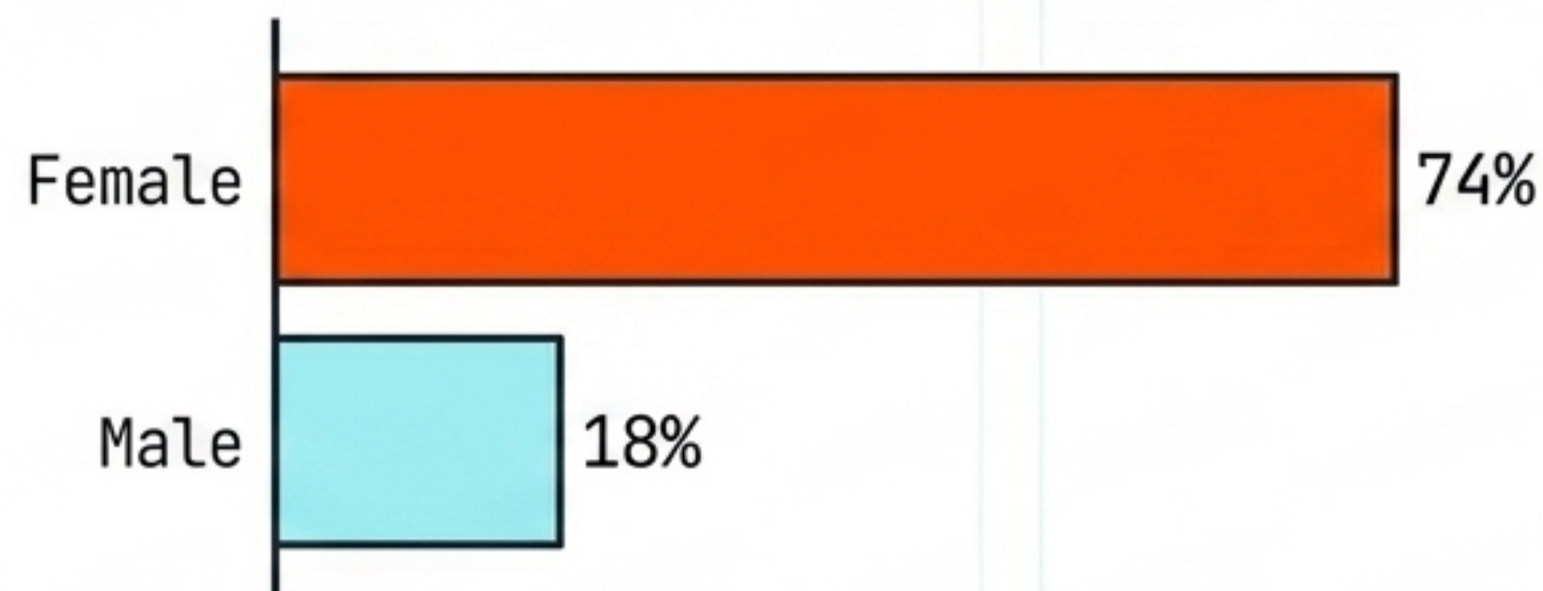


# The Necessity of Proportions

Heatmaps show volume, but volume can mislead if one category is simply larger than another. By grouping the data and calculating the mean of the binary Survived column, we extract the true survival percentage per category.

```
titanic.groupby('Pclass')['Survived'].mean() * 100
```

## Gender Survival Rate



## Class Survival Rate





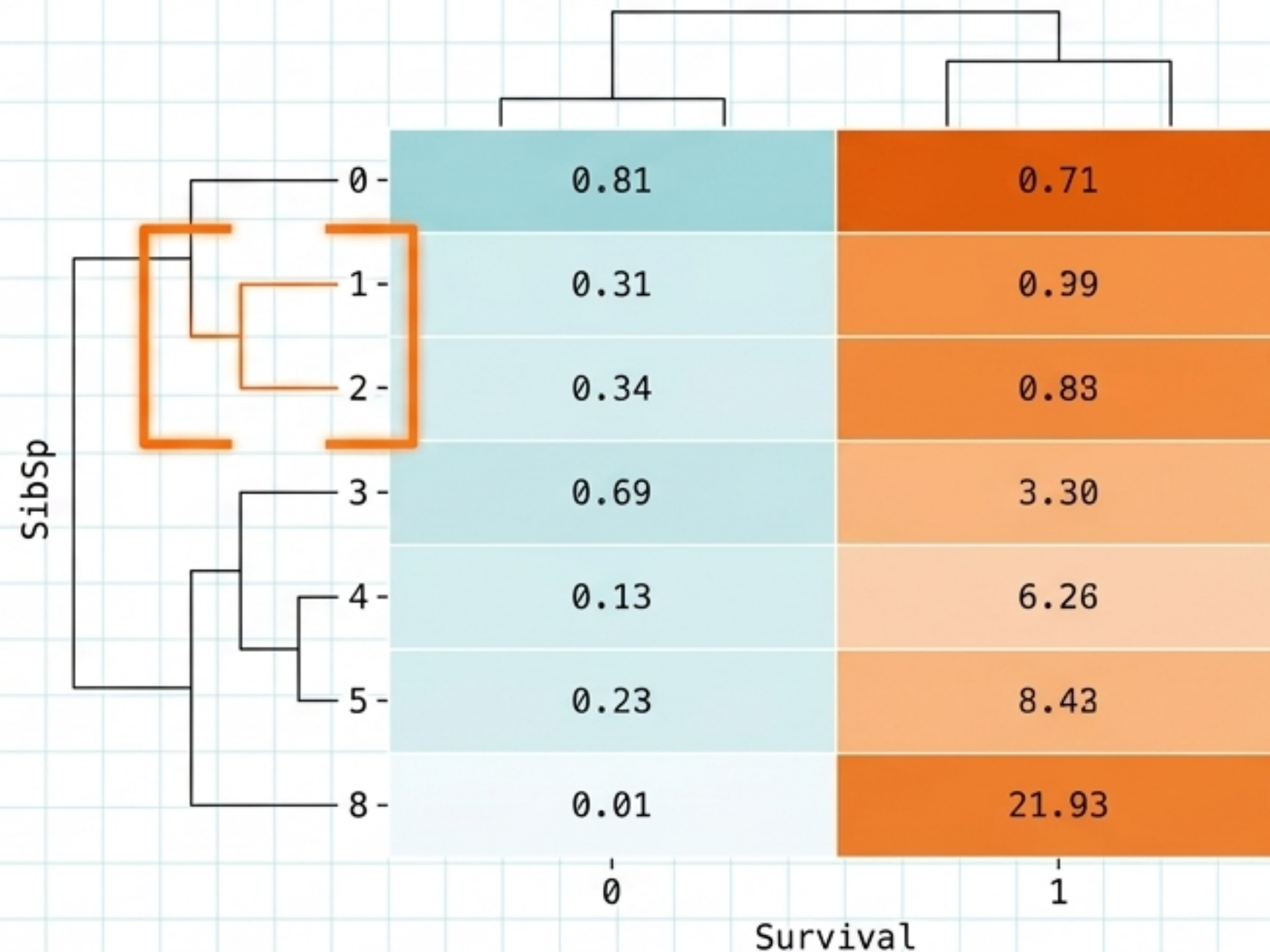
# Finding Hidden Relationships: Clustermaps

[CAT] × [CAT]

Clustermaps go beyond standard heat; they use hierarchical clustering (dendrograms) to group categories that behave similarly.

Insight: Passengers travelling with 1 or 2 companions share a nearly identical survival profile, which is distinctly different from lone travellers or large families.

`sns.clustermap`

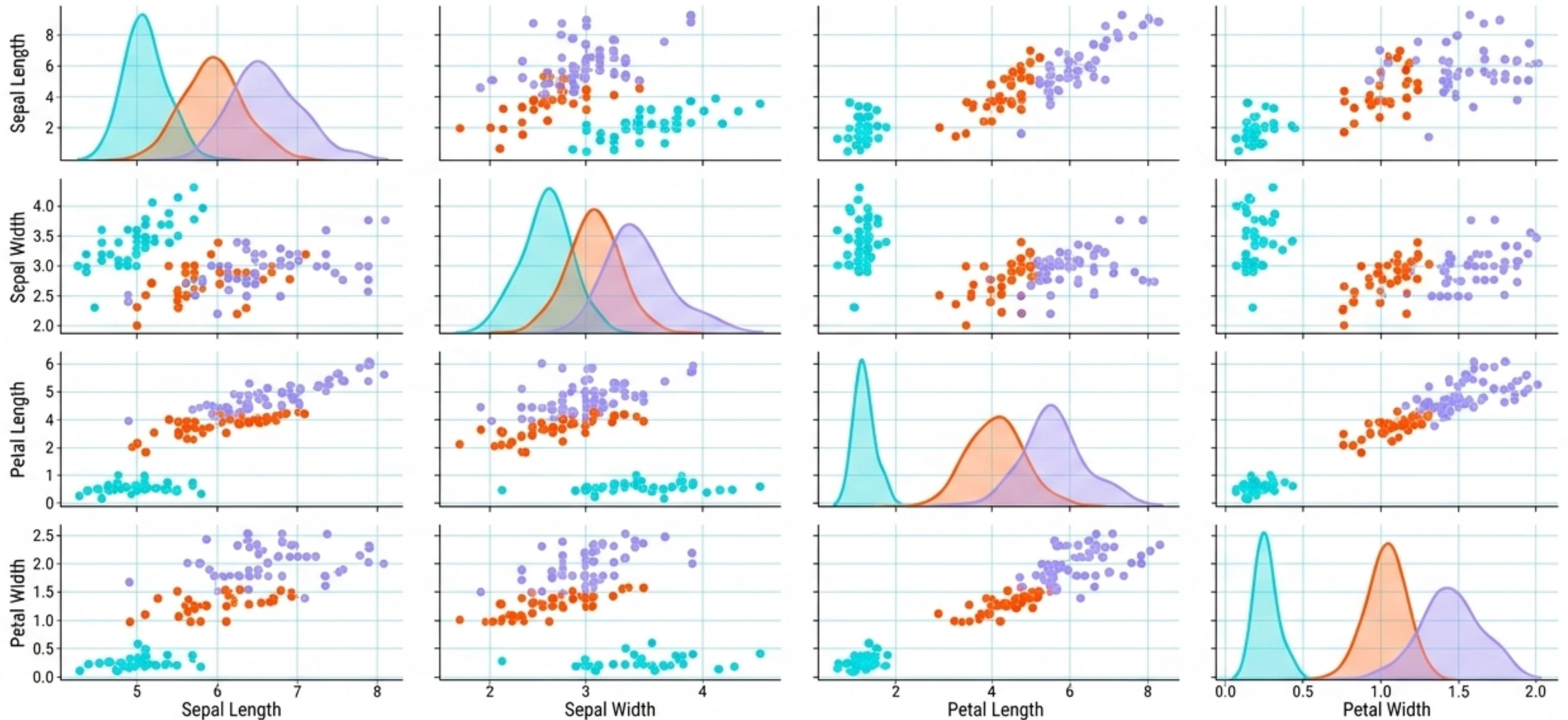




# The Macro View: Pairplots

[MACRO]

When facing multiple numerical columns, `sns.pairplot()` acts as a brute-force scatter matrix. It instantly maps every numerical combination against each other. Insight: Adding categorical colour encoding (hue) immediately reveals which physical dimensions best separate the three flower species.

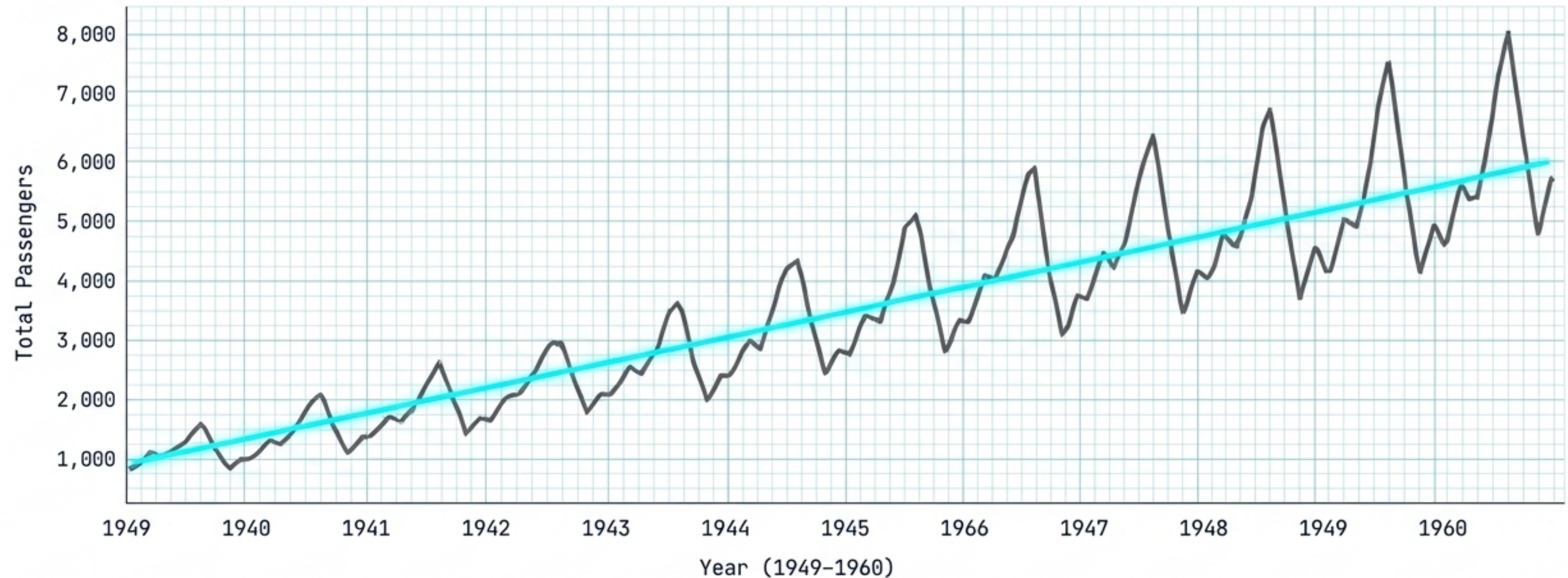




# Tracking Trajectories

[NUM] × [TIME]

Line plots are a highly specific variant of scatter plots where the points are connected. They are reserved exclusively for sequential, time-based data to expose long-term momentum and linear growth.



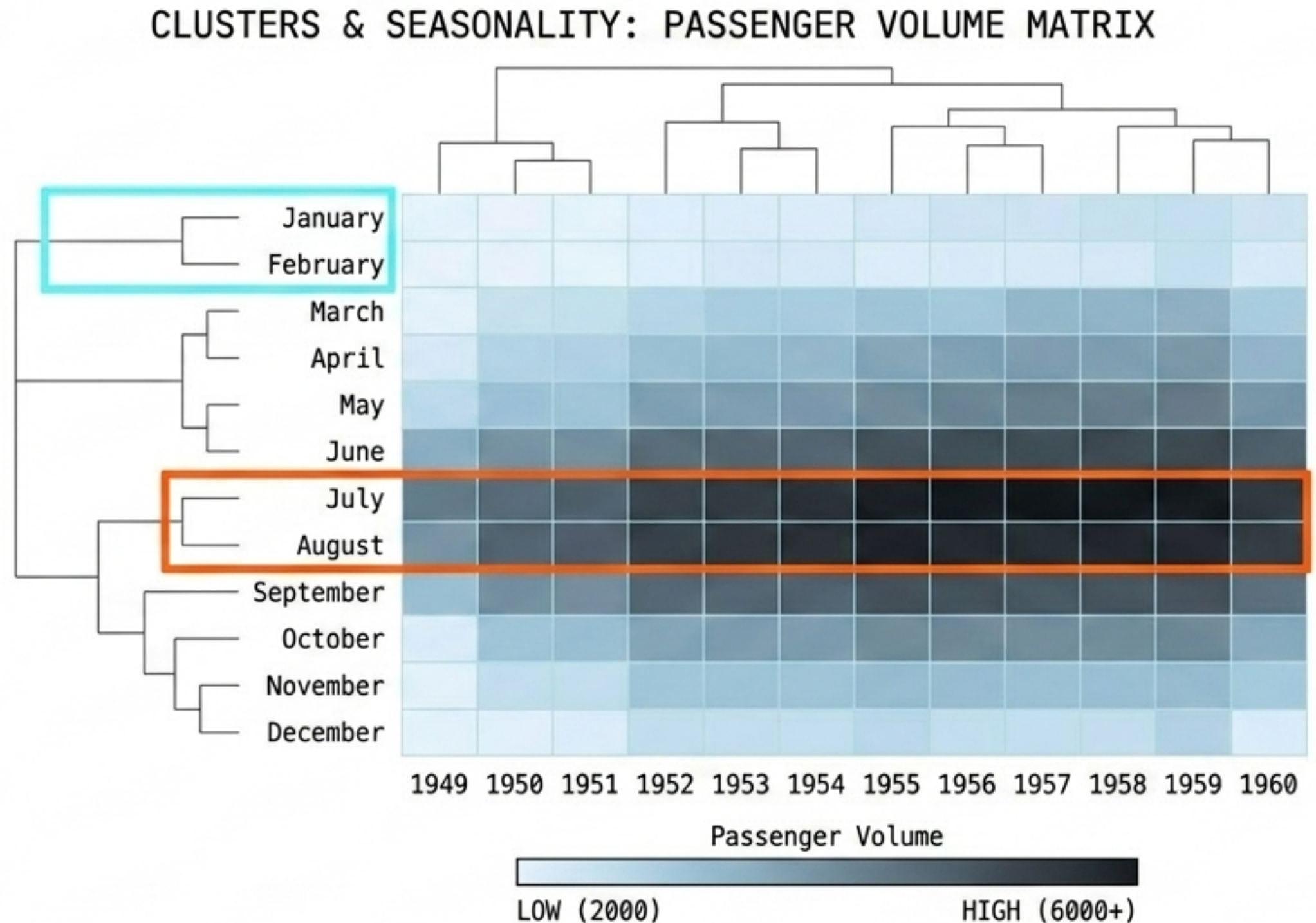


# Advanced Synthesis: Proving Seasonality

[TIME] × [SEASONALITY]

By reshaping time-series data into a Pivot Table and applying a **Clustermap**, seasonality is mathematically extracted.

**Insight:** The algorithm automatically groups **July** and **August** together as peak travel months across all years, while clustering January and February together as the absolute troughs.





# The EDA Visual Selection Matrix

Use this diagnostic matrix to match your Seaborn tool to your Pandas data types.

| DATA TYPE PAIRING (MONOSPACE) | SEABORN PLOT (SANS-SERIF)                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| [NUM] × [NUM]                 | Scatterplot (Find baseline correlations)<br>Pairplot (Macro scatter matrix)                                 |
| [NUM] × [CAT]                 | Barplot (Compare averages)<br>Boxplot (Find outliers & spread)<br>KDE/Distplot (Spot overlapping densities) |
| [CAT] × [CAT]                 | Heatmap (Spot intensity in crosstabs)<br>Clustermap (Find hierarchical groupings)                           |
| [NUM] × [TIME]                | Lineplot (Track sequential trajectory)                                                                      |



# The Analyst's Mindset

[END\_PROCESS]

**“EDA is not about writing code; it is about asking the right questions. The visualisations are simply the medium through which the data answers. Let the data tell the story.”**